STUDENT KEY
BIOLOGY 336 - Fundamentals of Evolutionary Biology W1
Midterm, October 21, 2025

1**.** In signing below**, please confirm that you agree to abide by an honour system** when taking this exam. You will not share your results with others or obtain information from others about any question. On our part, **we also abide by an honour system**, have attempted to write a fair and honest exam of your evolutionary knowledge, and will clarify any ambiguities.

2. Some questions ask that you show your work or include the formula that you used. Marks may be deducted for not doing this, and part marks may be given for showing your work.

3. If you think there may be a mistake in a question, if the meaning of a question is unclear, or if there is a word or phrase that you don't understand, please ask for clarification.

4**. If you think a question is ambiguous or open to interpretation, ask for a clarification and/or state your interpretation when writing the answer, as we will give credit for well-justified answers.**

5.The exam may be written in pen or pencil, but **pencil answers will not be eligible for regrading**.

6. Answer all questions in the space provided. Material written on other pages will not be read or marked unless you clearly indicate that you've completed the answer elsewhere.

7. You may use a non-programmable calculator (or show that your programmable calculator has nothing in memory).

8. Students who write the exam during the normally scheduled time may not discuss the contents of the exam with students who subsequently take a make-up version of the exam.

9. **Hand in all examination papers; do not take any examination material from the room.**

10. Please follow any additional examination rules or directions communicated by the instructor.

I have read and fully understand these instructions, and I have checked that all 16 pages (including extra page for rough work and the formula sheet) are present.

Student signature: _____

Page	Marks Possible	Your Mark
2.	9	
3.	4	
4.	4	
5.	6	
6.	10	
7.	7	
8.	14	
9	14	
10	10	
11	10	
12	6	
13	6	
TOTAL		

Q1: History of evolutionary thought [17 marks total]

(a) **[5 points. Circle the best response.]**

In 1859, Darwin's book "On the Origin of Species" was published for the first time. This work was the first to **[demonstrate that natural selection could account for the diversity of life on Earth / suggest that all life may descend from a single common ancestor].** Darwin's ideas were heavily influenced by the writings of **[Fisher / Lyell]**, who showed that **[the physical world changed gradually / selective breeding effected rapid changes]**, as well as the writings of **[Haldane / Malthus]**, who argued that **[most species are in competition for resources / all species eventually go extinct]**.

(b) **[2 points]**

One early criticism of Darwinism was from the engineer Fleeming Jenkin, who criticized the central logic of natural selection. Which of the following statements was most important for Jenkin's argument?

- □ Darwin's ideas challenged the story of creation told in the book of Genesis
- □ Jenkin's own experiments on fruitflies did not show evidence for change over time
- □ Blending inheritance prevented natural selection leading to novel phenotypes
- □ Most mutations are of large effect, refuting the gradual change suggested by Darwin

(c) **[2 points]**

During the first part of the 20th century, a number of researchers used mathematics to synthesize Mendelian genetics with the process of evolution. One result from this period is the equation for the per-generation change in allele frequency in haploids due to the action of natural selection:

Equation 1

$$\Delta p[t] = \frac{(W_A - W_a)p[t]q[t]}{W_A p[t] + W_a q[t]}$$

Central to Darwin's argument was the differential survival and reproduction of individuals within a population. Which term in Equation 1 captures differential survival and reproduction?

[we want WA-Wa, but give credit if they say it is the difference between the W's]

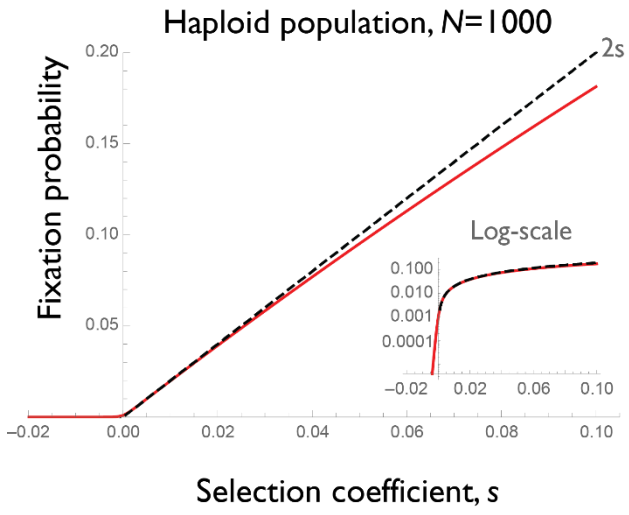
(d) **[2 marks]**

JBS Haldane used equations similar to equation 1 to derive an approximation for the probability that a beneficial allele will reach fixation, yielding his famous result that fixation probability is approximated by 2s in a haploid population. From Equation 1, what is the expression for 's'? State any assumptions that you make.

[Looking for (WA-Wa)/Wa. 1 mark if they just say what s is in words. 1 if they write WA-Wa and don''t state that they are assuming that Wa=1 (otherwise WA=2.2, WA=2 gives a different answer for s of 0.2 than WA=1.1, WA=1 with an s of 0.1, even though A is 10% more fit in both cases). We gave credit if s was implicitly defined by stating WA=1+s and Wa=1. In the literature, the reference used when calculating the probability of fixation of a new allele ("A") is always the resident allele ("a), but we also accepted answers where the reference used was "A", with WA=1 and Wa=1-s.]

(e) **[2 marks]**

In 1962, Kimura generalized Haldane's result. The figure inset shows a graph that compares Kimura's more precise formulation of fixation probability (solid), compared to Haldane's 1927 approximation (2s, dashed line). What is the biological significance of the fact that Kimura's result is positive to the left of the y-axis, whereas Haldane's is not?



[The more precise formulation shows that deleterious mutations (s<0) can, by chance, fix.]

(f) **[4 marks]**

The empirical basis for the Modern Synthesis came largely from work in macroorganisms. However, viruses often have much higher mutation rates. If mutation rates were on the same order of magnitude as selection coefficients, what other processes might Haldane need to include to derive an accurate approximation of the probability that a beneficial allele reaches fixation?

[Lots of reasonable things to write here, but the core logic is that if mutation rates are this high, then any allele is likely to mutate into a different allele before it reaches fixation, and the process is likely to involve multiple alleles that are identical through mutation, not just the emergence of a single rare allele. So Haldane's calculation would need to incorporate terms for forward/back mutation, and some assumptions about the mutational neighbourhood.

Any of the following points is worth a mark:

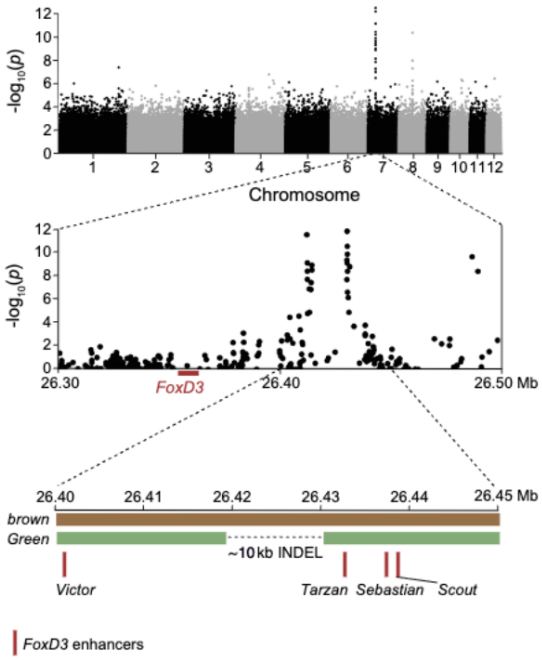
- The allele might mutate into a different allele
- If mutation is fast relative to selection then this might happen before fixation occurs
- We also need to know the chance that a different allele in the population mutates into the allele in question
- This means assuming something about the mutational spectrum
- Any reference to mutation/selection balance with respect to what 'fixation' means, e.g. fixation itself might not be a meaningful approximation, because mutation/selection balance might keep the allele far from 1

Q2 [51 points total]



Background: In a PNAS paper last month, Goutte and Boissinot (2025) examine colouration in frogs. Several species exhibit a polymorphism with green frogs and brown frogs, which is thought to enable animals to blend into their environments (against green grass or brown soils, respectively) and so reduce detection by predators.

The authors contrasted allele frequencies between green and brown frogs in the African grass frog species *Ptychadena robeensis*, illustrating their results in the "Manhattan plot" to the right. The Figure legend said: "Genome-wide association study of dorsal coloration in contrasting brown (n = 61) with green individuals (n = 39). The Green and brown alleles, as well as the known FoxD3 enhancers in the region are represented below the Manhattan plots."



(a) **[4 points]**

For each of the choices below, circle the best answer. The figure above shows statistical support for the [**green allele frequency / a genetic association with colour**] across the genome by chromosome. There are two main genes with a strong association, on chromosomes **[1&2 / 7&8**]. The first of these lies **[within / near]** the gene FoxD3, which is known in other species to affect specialized cells ("iridophores") that create structural colours (like green and blue) by reflecting light. The evidence shown in the bottom two panels leads to the conclusion that a change **[in the regulatory region / to the amino acid sequence]** of FoxD3 is associated with whether frogs of this species are green or brown.

(b) **[2 points]**

[True / False]: The existence of multiple black dots making up the chromosome 7 peak on the Manhattan plot proves that multiple genetic changes are required to switch colour.

(c) **[8 points]**

Linkage disequilibrium is high among the alleles segregating between positions 26.40 and 26.45 Mb. When looking at two of these sites, imagine that 61 chromosomes associated with brown all had nucleotides A and G at these two sites, while 39 chromosomes associated with green all had T and T at these two sites (no A-T or T-G chromosomes were found). What is the linkage disequilibrium between these two sites? [Write the result to four decimal points, e.g., 0.1111.]

FORMULA USED:

$$D[t] = x_{AB}[t]\, x_{ab}[t] - x_{Ab}[t]\, x_{aB}[t]$$

RESULT: Letting nucleotide A be the A at the first site and nucleotide G be the B allele at the second site, $x_{AB} = 61/100$, $x_{ab} = 39/100$, and $x_{Ab} = x_{aB} = 0$, so D = 0.2379. Different allele definitions could lead to D = –0.2379, which is fully acceptable too.

(d) **[2 points]**

Alleles within the FoxD3 gene itself do not display strong linkage disequilibrium with the sites associated with green/brown coloration. This is most likely because (select one):

□ Mutations do not occur in FoxD3

□ All mutations in FoxD3 change the amino acid sequence (replacement changes)

□ Recombination increases fitness of offspring

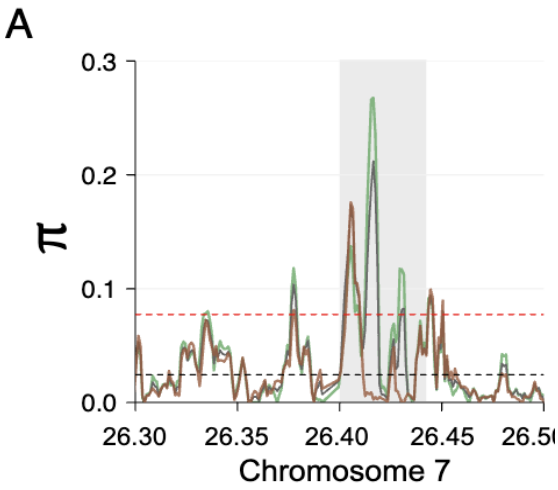
□ Recombination between these regions reduced linkage disequilibrium over time

□ The sites are completely linked ($r = 0$)

(e) **[7 points]**

The authors then looked at genetic diversity, measured by π (similar to the expected heterozygosity, *H*, that we discussed in class).

The top horizontal line to the right at 0.08 shows the genome-wide 95th percentile of nucleotide and the grey vertical box corresponds to the peak seen in the Manhattan plot above.



For each of the choices below, circle the best answer.

The figure shows higher [**allele frequency / expected heterozygosity]** in the region associated with brown versus green coloration in this frog species. This is consistent with **[directional selection / heterozygous advantage]**. The authors also compared genetic sequences from within this region, finding evidence that the sequence associated with brown *P. robeensis* frogs is more similar to the sequence associated with brown frogs from the related species *P. nana* than to the sequence associated with green *P. robeensis* frogs. This is an example of a **[trans-specific polymorphism / recurrent mutation]**, as seen also for humans at **[MHC immunity genes / disease-causing genes]**. While expected heterozygosity decreases, on average, by a factor **[$1 - \frac{1}{2N}$ / $1 - r$]** each generation due to **[genetic associations / random genetic drift]**, genetic diversity can be replenished by **[mutation / recombination]** and maintained by selection.

(f) **[8 points]**

If, across the normal habitat for this species, the fitnesses associated with the green (G) and brown (B) alleles were $W_{GG} = 1$, $W_{GB} = 1.2$, $W_{BB} = 0.8$, what would you expect the frequency of the green allele to be at equilibrium?

FORMULA USED: $\hat{p} = \frac{W_{Aa} - W_{aa}}{2W_{Aa} - W_{AA} - W_{aa}}$ [2points]

RESULT: $\hat{p} = \frac{1.2-0.8}{2*1.2-1-0.8} = 2/3$. Note that we have asked for the frequency of the green allele so A has to be green and W_{aa} has to be 0.8 [-3 points if they flip this and answer 1/3.]

(g) **[6 points]**

The authors state that "selection by visual predators is expected to impose a strong pressure on anuran [frog] cryptic color evolution," noting that "predators of these species likely include birds and mammals, which can form searching images" that allow them to detect frogs that do not match the background. In the Ethiopian highlands where *P. robeensis* frogs are found, climate warming and more frequent droughts are leading to a loss of green grass. Explain what you would expect to happen to the frequency of the green frog allele (G) over time as a result. **[Specify the direction of change (1 point) and explain why in 1-2 sentences (4 points), logic and structure (1 point)]**

[Predict that the frequency of the green frog allele should decline, because green frogs will not match the brown soil or brown dead grass. As predators will kill green frogs that don't match the background more often, the green allele should decline. A single clear sentence would be fine here. Something about predation or deaths should be said (-1 if not).]

(h) **[6 points]** Propose one experiment that would support your prediction. **[Briefly describe the design of the experiment and which outcome would support/fail to support your prediction (content: 5 points, logic and structure 1 point).]**

[Lots of possible answers: Expect more mortality of mis-matching frogs in experiments like:

- Release a mix of green and brown frogs in an open-air enclosure that has green grass versus dead brown grass, then return to measure survival.
- Put model frogs down and camera trap predation.
- Conduct a mark-release-recapture experiment in regions with mainly green habitat vs brown habitat.

(i) **[8 points]**

If, in a year of severe drought, the fitnesses of the frogs changed to $W_{GG} = 0.5$, $W_{GB} = 0.5$, $W_{BB} = 1$, what would the **change** in frequency of the green allele be from that generation to the next (Δp), assuming an initial green allele frequency of p=0.5?

FORMULA USED:

$$p[t+1] = \frac{W_{AA}\, p[t]^2 + W_{Aa} p[t] q[t]}{W_{AA}\, p[t]^2 + W_{Aa} 2\, p[t] q[t] + W_{aa}\, q[t]^2}$$

RESULT:

$$p[t+1] = \frac{0.5 * 0.5^2 + 0.5 * 0.5 * 0.5}{0.5 * 0.5^2 + 2 * 0.5 * 0.5 * 0.5 + 1 * 0.5^2} = 0.4$$

So $\Delta p = -0.1$. (–1 for each simple algebraic typo, -2 for giving p[t+1] but not the change in allele frequency)

Q3 [12 marks total]

In humans, the same gene discussed above (FoxD3) was found "in affected members of a large family with autosomal dominant vitiligo," which is a chronic autoimmune disorder that causes patches of skin to lose pigment (Alkhateeb et al. 2005). The authors used a similar pedigree analysis as did Dr. Mary Claire King to find the BRCA1 gene underlying breast cancer. In this case, Alkhateeb et al. (2005) identified a single mutation in the promoter of FoxD3 that altered expression in some "permissive" cells (neural crest melanoblast precursors). The incidence of vitiligo is very high (~0.4% of people, Bahrt Haulrig et al. 2024), including many famous people such as Michael Jackson and Canadian fashion model Winnie Harlow (a public spokesperson for the condition).

In heterozygotes, vitiligo does not increase mortality*, but homozygotes for these mutations are never seen. Here we assume that the skin condition (vitiligo) is autosomal dominant, but that it is recessive with respect to mortality, with all homozygotes dying.

Fill in the blanks with the best answer from this list: h, s, $2pq$, q^2, W_{AA}, W_{Aa}, W_{aa}, $\frac{\mu}{hs}$, $2\frac{\mu}{hs}$, $\sqrt{\frac{\mu}{s}}$, $\frac{\mu}{s}$.

[1 point each] This is an example where dominance depends on the trait considered. The fact that vitiligo, in heterozygotes, does not increase mortality* suggests that _____ W_{Aa}_____ is the same as the fitness of individuals without vitiligo (W_{AA}). When focusing on fitness, the phrase "recessive with respect to mortality" indicates that ___h_____ is zero, and the fact that homozygotes for these mutations are never found suggests that ___s_____ is one. At mutation-selection balance, the frequency of the mutant allele would thus be expected to equal_$\sqrt{\frac{\mu}{s}}$_____. The number "0.4%" represents the frequency of individuals with the skin condition (seen in heterozygotes), which is expected to be __$2pq$_____ in a randomly mating population.

[5 points] Putting the above information together, an estimate of the mutation rate is (**X** the □ for your answer):

- □ $\frac{0.004}{2}$ 1
- □ $\frac{0.004}{2}$ 0
- □ (0.004) 1
- □ (0.004) 1
- □ None of the above
- □ $\left(\frac{0.004}{2}\right)^2$ 1
- □ $\left(\frac{0.004}{2}\right)^2$ 0
- □ $(0.004)^2$ 1
- □ $(0.004)^2$ 0

* A recent study of 107,424 patients with vitiligo and 537,120 matched controls from South Korea found that mortality rates were 25% lower among individuals with vitiligo (Ju et al. 2024), with lower risks of dying from infectious diseases, cancer, and several other causes of mortality, perhaps due to the altered immune response.

[2 points. Circle one: True / False] This new finding suggests that vitiligo might be common not because the mutation rate is high but because of heterozygous advantage.

Q4 Speciation [20 marks]

(a) **[6 marks]**

In the 20th century, many leading researchers claimed that sympatric speciation was virtually impossible in nature and should therefore be rare or even nonexistent. Do you agree? With reference to a real example, why/why not? Your answer define sympatry/allopatry and mention its influence on the speciation process **(content 5 points, structure 1 point)**.

[1 mark 'no'; 1 for correct definition; another for any example of sympatric speciation given; two marks for a correct explanation of how speciation forms in this case; one mark for any reasonable statement about how general it might be, i.e. getting at 'rare']

[1 mark 'yes'; 1 for correct definition; 1 point: should acknowledge that some cases of sympatric speciation are known, but they are still likely to be rare relative to cases where there is geographic separation (allopatry) during speciation; two marks for a correct explanation of how allopatry/sympatry affects speciation; one mark for any reasonable statement about how general sympatric speciation might be, i.e. getting at 'rare']

(c) **[2 marks]**

In the late 19th and early 20th centuries, 'mutationists' such as Hugo de Vries and Richard Goldschmidt argued that new species were more likely to arise from single, large-effect mutational events, than from the kind of gradual accumulated change envisaged by Darwin. Would the mutationists' conception of speciation be closer to the modern notion of sympatric or allopatric speciation, and why? [One sentence]

[1 mark for sympatric; 1 for stating that physical separation not required here]

(d) **[3 marks]**

Choose an empirical example where mechanisms of speciation have been studied and argue whether the evidence in that case is more consistent with a single large-effect mutational event (as envisaged by mutationists) or the accumulation of gradual changes over time (as envisaged by Darwin).

[1 mark for a good example; 1 for correctly stating if it's rapid mutational event (e.g. polyploid speciation) or 'gradualist' (e.g. divergence in greenish warblers); 1 for a good argument why

Note most examples should be gradualist]

(e) **[3 points]**

Polyploidization is often argued to be a common route to speciation in plants. Scientists have argued that this may be due to incompatibilities between polyploids and diploids (when polyploid and diploids mate, they produce infertile triploids). Others have focused on phenotypic differences, as polyploids often differ in seed size, development, and various physiological traits. At a particular geographical location with both diploid and polyploid individuals present, name one species concept **(1 point)** and describe one piece of information that you would most want to know to determine whether the polyploids are the same or a different species from the diploid plants based on that species concept **(2 points).**

Lots of potential answers (see Jeannette's slide on species concepts):

- For the genotypic cluster species concept, may want to sequence and look at genetic association (persistent linkage disequilibrium)
- For the biological species concept, may want to confirm that triploid fitness is near zero.
- For the ecological species concept, may want to confirm that diploids have higher fitness under different ecological conditions than polyploids.

(f) **[6 marks]** Sometimes, 'good species' can produce hybrid offspring. In 1922, JBS Haldane made an observation now known as "Haldane's Rule", which states that: "When in the F1 offspring of two different species one sex is absent, rare, or sterile, that sex is the heterozygous sex (heterogametic sex)". We now know that Haldane's Rule holds broadly across multiple species, including mammals (in which males are heterogametic with XY chromosomes, while females are XX), and butterflies (in which females are heterogametic with ZW chromosomes, while males are ZZ). Based on your understanding of population genetics, what might explain Haldane's Rule? In 3-4 sentences, suggest one factor that might explain why, explain your reasoning in words or with a brief diagram, and suggest an experiment or line of empirical evidence that could support or refute your proposed factor **(5 points content, 1 point structure)**.

[1 point for any reasonable idea, up to a further 2 points for logic supporting the idea, 1 point for a reasonable experiment, 1 point for logic connecting the experimental outcomes to the idea, and 1 point for overall structure of the answer]

Credit was given for any reasonable answer, including any of the following table, which reviews arguments for Haldane's Rule that exist in the literature, taken from Cowell 2023 https://academic.oup.com/jeb/article/36/2/337/7326090 :

Theory	Applicable to all sex determination systems?	Genetic explanation	Source
Sexual Transformation	Yes	Hybrids are transformed into the homogametic sex but remain chromosomally heterogametic.	Haldane (1922); Goldschmidt (1934)
Chromosomal Rearrangements	Yes	A piece of chromosome breaks off the Y or W chromosome and is relocated on another, causing an incompatibility.	Haldane (1932)
Y Incompatibilities	Yes	Y is incompatible with X chromosome or autosomes.	Muller (1942)
Dosage Compensation	Yes	Hybridisation causes problems with regulation and/or activation of the dosage compensation process needed in heterogametes.	Coyne and Orr (1989b)
Dominance	No [Because marsupial females inactivate one X]	Recessive X- or Z-linked genes cause sterility.	Muller (1942)
Faster-Male	No	Faster divergence of male reproductive genes than female ones.	Wu and Davis (1993)
Faster-X	Yes (but not a direct cause)	Faster divergence of X- or Z-linked loci relative to autosomal ones.	Charlesworth et al. (1987)
Meiotic Drive	Yes	Discrepancy between drivers and suppressors on sex chromosomes leads to sterility.	Frank (1991); Hurst and Pomiankowski (1991)

Extra page for rough work

Biology 336 Formula Sheet – Midterm Exam Version

Haploid selection:

$$p[t+1] = \frac{W_A\, p[t]}{W_A p[t] + W_a q[t]}$$

$$\Delta p[t] = p[t+1] - p[t] = \frac{(W_A - W_a)\, p[t]\, q[t]}{W_A\, p[t] + W_a\, q[t]}$$

Allele frequency after several generations:

$$\frac{p[t]}{q[t]} = \frac{W_A^t\, p[0]}{W_a^t\, q[0]}$$ or equivalently $$p[t] = \frac{W_A^t\, p[0]}{W_A^t\, p[0] + W_a^t\, q[0]}$$

Mean fitness:

$$\bar{W} = W_A p[t] + W_a q[t]$$

Diploid selection:

Genotype frequencies when in "Hardy-Weinberg proportions":

$$x_{AA}[t] = p[t]^2, x_{Aa}[t] = 2p[t]q[t], x_{aa}[t] = q[t]^2$$

Genotype frequencies after selection

$$x_{AA}^s = \frac{W_{AA} x_{AA}[t]}{\bar{W}}, x_{Aa}^s = \frac{W_{Aa} x_{Aa}[t]}{\bar{W}}, x_{aa}^s = \frac{W_{aa} x_{aa}[t]}{\bar{W}}$$

where the mean fitness is:

$$\bar{W} = W_{AA} x_{AA}[t] + W_{Aa} x_{Aa}[t] + W_{aa} x_{aa}[t]$$

Allele frequency in the next generation

$$p[t+1] = \frac{W_{AA}\, p[t]^2 + W_{Aa} p[t] q[t]}{W_{AA}\, p[t]^2 + W_{Aa} 2\, p[t] q[t] + W_{aa}\, q[t]^2}$$

Formula for polymorphic equilibrium (used only for Heterozygote advantage or disadvantage):

$$\hat{p} = \frac{W_{Aa} - W_{aa}}{2W_{Aa} - W_{AA} - W_{aa}}$$

Mutations:

Mutation alone (no selection, haploid or diploid model): $p[t+1] = (1-\mu)\, p[t] + \nu\,(1 - p[t])$

Equilibrium: $\hat{p} = \frac{\nu}{\mu + \nu}$

Haploid model: with mutations and selection ($W_A = 1, W_a = 1 - s$)*, the deleterious allele "a" approaches:* $\hat{q} \approx \frac{\mu}{s}$

Diploid model: with mutations and selection, the deleterious allele "a" approaches:

$\hat{q} \approx \frac{\mu}{hs}$ *if h>0 and* $\hat{q} \approx \sqrt{\frac{\mu}{s}}$ *if h=0 (i.e., if "a" is recessive to "A")*

At the diploid mutation-selection balance equilibrium ($\hat{q} \approx \frac{\mu}{hs}$), the mean fitness is $\bar{W} = 1 - 2\mu$, which does not depend on the strength of selection.

Genetic Drift:

- The chance that a neutral allele becomes fixed is its initial frequency *p*.
- The time for a neutral allele initially in a single copy to become fixed: ~2*N* generations in a haploid population; 4*N* generations in a diploid population.

Expected heterozygosity, $H[t]$, the chance that two randomly chosen alleles differ, is a measure of genetic variation.

$$H[0] = 2pq$$

$$H[t] = \left(1 - \frac{1}{2N}\right) H[t-1]$$

Expected homozygosity, $F[t]$, is the chance that two randomly chosen alleles are the same, $F[t] = 1 - H[t]$.

Haldane's fixation probability: 2*s* in haploids and *2hs* in a diploid (assuming *h*>0).

Two-locus model:

Disequilibrium, D (also known as linkage disequilibrium or gametic-phase disequilibrium), measures genetic associations among two loci.

$$D[t] = x_{AB}[t]\, x_{ab}[t] - x_{Ab}[t]\, x_{aB}[t]$$

Recombination acts to break down linkage disequilibrium. Meiosis with recombination changes the chromosome frequencies by:

$$x_{AB}^{M} = x_{AB} - r\, D$$

$$x_{Ab}^{M} = x_{Ab} + r\, D$$

$$x_{aB}^{M} = x_{aB} + r\, D$$

$$x_{ab}^{M} = x_{ab} - r\, D$$

When selection is absent (all $W_{ij} = 1$), genetic associations (D) decay by a factor $(1 - r)$ each generation:

$$D[t] = (1 - r) D[t-1]$$

This means that over a number of generations, t, genetic associations decay according to:

$$D[t] = (1 - r)^t\, D[0]$$